

Final Review: Key Ideas/Techniques

Chapter 6: Review

Key Ideas/Techniques

1. Definition of upper and lower (Darboux) sums.
2. Definition of $\int_a^b f$ in terms of the upper and lower (Darboux) integrals.
3. Cauchy criterion for integrability (definition in terms of ϵ, P).
4. Monotone $\Rightarrow$ Integrable
5. Continuous $\Rightarrow$ Integrable

Activity

A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is integrable if for every $\epsilon > 0$ there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$.

Prove, using the Cauchy criterion for integrability (ϵ, P definition) that a real-valued function that is defined and constant on the interval $[a, b]$, except for at some point $x_0 \in (a, b)$, is integrable.

Define $f(x) = \begin{cases} M & x \in (a, b), x \neq x_0 \\ c & x = x_0 \end{cases}$ f is bounded since $|f(x)| \leq \max\{M, c\}$.
Let $\epsilon > 0$. Define $P = \{a, t_1, t_2, b\}$ such that $x_0 \in [t_1, t_2]$

$$\text{and } |t_2 - t_1| < \frac{\epsilon}{|M - c|}.$$

CASE 1 - $M > c$: Assume $M > c$, then

$$U(f, P) - L(f, P) = (M - M)(t_1 - a) + (M - c)(t_2 - t_1) + (M - M)(b - t_2) = (M - c)(t_2 - t_1) < \epsilon$$

CASE 2 - $M < c$: Assume $M < c$, then

$$U(f, P) - L(f, P) = (M - M)(t_1 - a) + (c - M)(t_2 - t_1) + (M - M)(b - t_2) = (c - M)(t_2 - t_1) < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, by the Cauchy criterion for integrability, $f(x)$ is integrable on $[a, b]$.

Chapter 5: Review

Key Ideas/Techniques

1. Limit definition of the derivative.
2. Know how to compute $f'(x)$ using the definition.
3. Algebraic properties of the derivative (sum, product, quotient, chain rule)
4. Differentiable $\Rightarrow$ Continuous
5. Mean Value Theorem and applications

Activity

If $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists and is finite

then $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

Compute f' if $f(x) = \frac{1}{x}$ using the limit definition.

$$\lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a - x}{xa}}{x - a} = \lim_{x \rightarrow a} \frac{-1}{xa} = \frac{-1}{a^2}.$$

Thus $f'(a) = \frac{-1}{a^2}.$

Relating functions & derivatives $\rightarrow$ MVT & corollaries.

Activity

Suppose $f, g: \mathbb{R} \rightarrow \mathbb{R}$ are differentiable. Prove if $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$, then $f(x) \leq g(x)$ for all $x \geq 0$.

Hint: Consider $h = g - f$.

Let $h(x) = g(x) - f(x)$. Then $h'(x) = g'(x) - f'(x) \geq 0$. So h is increasing. So if $x \geq 0$, then $h(x) \geq h(0) = g(0) - f(0) = 0$.

thus $g(x) - f(x) \geq 0$ and $g(x) \geq f(x)$ for all $x \geq 0$.

Activity

Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $f': \mathbb{R} \rightarrow \mathbb{R}$ is also differentiable. Prove if $f(0) = f'(0) = 0$ and $f''(x) \leq M$ for all $x \geq 0$, then $f(x) \leq \frac{M}{2}x^2$ for all $x \geq 0$.

Hint: Apply the previous activity twice.

Let $h(x) = f(x) - \frac{M}{2}x^2$. Then $h'(x) = f'(x) - Mx$ and $h''(x) = f''(x) - M$.

Then $h''(x) \leq 0$ for all $x \geq 0$ so h' is decreasing on $[0, \infty)$. Thus, if $x \geq 0$ $h'(x) \leq h'(0) = f'(0) - M \cdot 0 = 0 - 0 = 0$. so h is decreasing on $[0, \infty)$.

So if $x \geq 0$, then $h(x) \leq h(0) = f(0) - \frac{1}{2}M \cdot 0^2 = 0 - 0 = 0$. Thus $f(x) - \frac{M}{2}x^2 \leq 0$

and $f(x) \leq \frac{M}{2}x^2$ for all $x \geq 0$.

Activity

Suppose $f: (a, b) \rightarrow \mathbb{R}$ is differentiable and $f': (a, b) \rightarrow \mathbb{R}$ is also differentiable. Prove if $f(x_1) = f(x_2) = f(x_3) = 0$ for some $x_1 < x_2 < x_3$, then $f''(c) = 0$ for some $c \in \mathbb{R}$.

Hint: Apply the MVT to f on the intervals $[x_1, x_2]$ and $[x_2, x_3]$.

By the MVT there exist $y_1 \in (x_1, x_2)$ and $y_2 \in (x_2, x_3)$ such that

$$f'(y_1) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{0 - 0}{x_2 - x_1} = 0 \quad \text{and} \quad f'(y_2) = \frac{f(x_3) - f(x_2)}{x_3 - x_2} = \frac{0 - 0}{x_3 - x_2} = 0.$$

Since f' is also differentiable we can apply the MVT to $[y_1, y_2]$, so there exists some $c \in (y_1, y_2)$ such that

$$f''(c) = \frac{f'(y_2) - f'(y_1)}{y_2 - y_1} = \frac{0 - 0}{y_2 - y_1} = 0.$$

Note that $y_1 \neq y_2$ since $y_1 \in (x_1, x_2)$ and $y_2 \in (x_2, x_3)$.

Activity

Prove that the equation $x^3 - 3x + 1 = 0$ has exactly three solutions in $\mathbb{R}$.

show $\begin{matrix} \leq \\ \text{at most} \\ \hline \text{at least} \\ \hline \geq \end{matrix} \begin{matrix} 3 \\ 3 \end{matrix}$

Hint: Where is the function $f(x) = x^3 - 3x + 1$ increasing/decreasing?

Note that $f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x-1)(x+1)$. So $f'(x) = 0$ $x = -1, 1$ and $f'(x) > 0$ on $(-\infty, -1) \cup (1, \infty)$ and $f'(x) < 0$ on $(-1, 1)$.

So f is strictly increasing on $(-\infty, -1]$ and $[1, \infty)$ and f is strictly decreasing on $[-1, 1]$. Then f has at most one solution on each interval $\Rightarrow f(x) = 0$ has at most 3 solutions.

Observe that $f(-1) = -1 + 3 + 1 = 3$, $f(1) = 1 - 3 + 1 = -2$, $f(-2) = -8 + 6 + 1 = -1$, and $f(2) = 8 - 6 + 1 = 3$ so by the IVT, there exist at least one solution on each interval $(-2, -1)$, $(-1, 1)$, $(1, 2)$. $\Rightarrow f(x) = 0$ has at least 3 solutions. Thus $f(x) = 0$ has exactly 3 solutions.

Activity

Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable. Prove if

- $|f(x)| \leq 1$ and
- $f'(x) > -1$ for all $x \in \mathbb{R}$,

then the equation $f(x) + x = 0$ has exactly one solution in $\mathbb{R}$.

Let $g(x) = f(x) + x$. Then $g'(x) = f'(x) + 1 > -1 + 1 = 0$. So g is strictly increasing on $\mathbb{R}$, so $g(x) = 0$ has at most one solution.

Note $|f(x)| \leq 1$, so $-1 \leq f(x) \leq 1$, and $-1 + x \leq g(x) \leq 1 + x$. Since $f(x)$ and x are differentiable, g is differentiable and therefore continuous. Note that $g(-2) \leq 1 + (-2) = -1 < 0$ and $g(2) \geq -1 + 2 = 1 > 0$. So by the IVT, there exists some $x_0 \in (-2, 2)$ with $g(x_0) = 0$. So $g(x) = 0$ has at least one solution. Therefore $f(x) + x = 0$ has exactly one solution.

Activity

Show that the sum of a positive number and its reciprocal is at least 2, i.e. show that $f(x) = x + \frac{1}{x}$ is bounded below by 2 on $(0, \infty)$.

Hint: Determine where f is increasing and decreasing.

Let $f(x) = x + \frac{1}{x}$. Then $f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}$. So $f'(x) < 0$ on $(0, 1)$ and $f'(x) > 0$ on $(1, \infty)$. So f is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$. This implies f has a minimum at $x = 1$. So

$$f(x) \geq f(1) = 1 + \frac{1}{1} = 2.$$

for all $x \in (0, \infty)$.

Activity

Suppose $f: (0, \infty) \rightarrow \mathbb{R}$ is a function such that $f'(x) = \frac{1}{x}$ for all $x > 0$ and $f(1) = 0$. Prove $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$.

Hint: Fix y and compute the derivative of $f(xy)$ and $f(x) + f(y)$.

Note: $f(x) = \log(x)$, but we haven't studied this function. So don't use it.

Fix $y > 0$. Set $g(x) = f(xy)$ and $h(x) = f(x) + f(y)$. Then

$$g'(x) = f'(xy) \cdot y = \frac{1}{xy} \cdot y = \frac{1}{x} \quad \text{and} \quad h'(x) = f'(x) + 0 = \frac{1}{x}.$$

So $g' = h'$ on $(0, \infty)$. Thus there exists a constant C such

that $g(x) = h(x) + C$. $g(1) = f(y)$ and $h(1) = f(1) + f(y) = f(y)$ so

$C = 0$. Thus $g(x) = h(x)$ for all $x > 0$. Since $y > 0$ was arbitrary,

$$f(xy) = f(x) + f(y) \quad \text{for all } x, y > 0.$$

Chapter 3: Review

Key Ideas/Techniques – Limits

1. Definition of $\lim_{x \rightarrow a} f(x)$ using ϵ, δ and the proof template for verifying it.
2. Equivalent definition of $\lim_{x \rightarrow a} f(x)$ using sequences.
3. Algebraic properties of limits (sums, products, quotients)
4. Template for showing $\lim_{x \rightarrow a} f(x)$ does not exist
 - Find sequences (x_n) and (y_n) in $\text{dom}(f) - \{a\}$ where $\lim x_n = a = \lim y_n$, but $\lim f(x_n) \neq \lim f(y_n)$.

Activity

Suppose $\text{dom}(f) = \mathbb{R}$ and $\lim_{x \rightarrow 3} f(x) = L$. Prove using the ϵ, δ definition that $\lim_{x \rightarrow 1} f(2x + 1) = L$.

Template: For proving $\lim_{x \rightarrow a} f(x) = L$

- “Fix $\epsilon > 0$.”
- “Let $\delta = ?$ ”
- “Assume $0 < |x - a| < \delta$.”
- Show $|f(x) - L| < \epsilon$.
- “Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow a} f(x) = L$.”

Fix $\epsilon > 0$. Since $\lim_{x \rightarrow 3} f(x) = L$,
there exists $\delta_1 > 0$ such that if
 $|x - 3| < \delta_1$, then $|f(x) - L| < \epsilon$.
Set $\delta = \frac{\delta_1}{2}$. Assume $|x - 1| < \delta$

$$|(2x + 1) - 3| = |2x - 2| = 2|x - 1| < 2 \cdot \frac{\delta_1}{2} = \delta_1$$

$$\text{so } |f(2x + 1) - L| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary,

$$\lim_{x \rightarrow 1} f(2x + 1) = L.$$

Activity

Let

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Prove if $a \in \mathbb{R}$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

Template: Find sequences (x_n) and (y_n) in $\text{dom}(f) - \{a\}$ where $\lim x_n = a = \lim y_n$, but $\lim f(x_n) \neq \lim f(y_n)$.

Fix $a \in \mathbb{R}$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, for each $n \in \mathbb{N}$ there exists $x_n \in \mathbb{Q}$ with $a < x_n < a + \frac{1}{n}$. Let $y_n = x_n + \frac{\sqrt{2}}{n}$. Then y_n is irrational (since $\sqrt{2}$ is irrational, as proven in class). Then $\lim x_n = a = \lim y_n$ but $\lim f(x_n) = 1$ and $\lim f(y_n) = 0$. So $\lim_{x \rightarrow a} f(x)$ does not exist.

Key Ideas/Techniques – Continuity

1. Definition of continuity using sequences.
2. Equivalent definition using ϵ , δ and the proof template for verifying it.
3. Algebraic properties of continuity (sums, products, quotients, composition)
4. Template for showing a function is not continuous at a point.

Activity

Suppose $f, g: S \rightarrow \mathbb{R}$ are continuous at x_0 . Prove using the ϵ, δ definition that $f + g$ is continuous at x_0 .

Fix $\epsilon > 0$. Since f is continuous at x_0 there exists $\delta_1 > 0$ such that:

if $x \in S$ and $|x - x_0| < \delta_1$, then $|f(x) - f(x_0)| < \frac{\epsilon}{2}$. Since g is

continuous at x_0 there exists $\delta_2 > 0$ such that if $x \in S$ and

$|x - x_0| < \delta_2$, then $|g(x) - g(x_0)| < \frac{\epsilon}{2}$. Let $\delta = \min\{\delta_1, \delta_2\}$. Assume

$x \in S$ and $|x - x_0| < \delta$. Then

$$|f(x) + g(x) - (f(x_0) + g(x_0))| = |f(x) - f(x_0) + g(x) - g(x_0)| \leq$$

$$|f(x) - f(x_0)| + |g(x) - g(x_0)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \text{ Since } \epsilon > 0$$

was arbitrary, $f + g$ is continuous at x_0 .

Activity

Show that

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is not continuous at 0.

Template: Find a sequence (x_n) in $\text{dom}(f)$ where $\lim x_n = 0$, but $\lim f(x_n) \neq f(0)$.

$$\text{Let } x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}$$

$$\lim x_n = 0$$

$$\lim f(x_n) = \lim \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1 \neq f(0).$$

Key Ideas/Techniques – Continuity

1. Maximums and minimums exist for continuous functions on closed intervals.
2. Intermediate Value Theorem.
3. Using the Intermediate Value Theorem in applications.
4. Definition of uniform continuity and the proof template for verifying it.

Recipe for Solving IVT Problems

Theorem. (IVT) Let f be a continuous real-valued function on a closed interval $[a, b]$. If y is a value between $f(a)$ and $f(b)$, then there exists at least one $x \in (a, b)$ such that $f(x) = y$.

- I. Define a function $f(x)$.
- II. Define a number y .
- III. Choose an interval $[a, b]$.
- IV. Establish that f is continuous on $[a, b]$.
- V. Establish that y is between $f(a)$ and $f(b)$.
- VI. Apply the conclusion of the IVT.

Activity

Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $|f(x)| \leq 1$ for all $x \in \mathbb{R}$. Prove that there exists at least one point x_0 such that $f(x) = 6 - x^3$.

Template: for applying the IVT

- I. Define a function $f(x)$.
- II. Define a number y .
- III. Choose an interval $[a, b]$.
- IV. Establish that f is continuous on $[a, b]$.
- V. Establish that y is between $f(a)$ and $f(b)$.
- VI. Apply the conclusion of the IVT.

Let $g(x) = f(x) + x^3$. Then g is continuous on $\mathbb{R}$. Since

$|f(x)| \leq 1$ for all x we have

$$-1 + x^3 \leq g(x) \leq 1 + x^3$$

$$g(0) \leq 1 + 0 = 1 < 6$$

$$g(2) \geq -1 + 2^3 = 7 > 6$$

So, by the IVT, there exists some $x_0 \in (0, 2)$

with $g(x_0) = 6$. Then

$$f(x_0) = 6 - x_0^3.$$

Activity

Prove that if $a \geq 0$, then the equation $x^2 = a$ has a solution.

Activity

Suppose $f, g: S \rightarrow \mathbb{R}$ are uniformly continuous on S . Prove that $f + g$ is uniformly continuous on S .

Template: To show that $f: S \rightarrow \mathbb{R}$ is uniformly continuous on S .

- I. “Fix $\epsilon > 0$.”
- II. “Let $\delta = (\textit{something})$.” [this requires scratch work]
- III. “Assume $x, y \in S$ and $|x - y| < \delta$.”
- IV. Show that $|f(x) - f(y)| < \epsilon$.
- V. “Since $\epsilon > 0$ was arbitrary, f is uniformly continuous on S .”

Chapter 2: Review

Key Ideas/Techniques – Sequences

1. Definition of $\lim s_n = L$ and the proof template for verifying it.
2. Algebraic properties of limits (sums, products, quotients)
3. Definition of $\lim s_n = \pm\infty$ and the proof template for verifying it.
4. Definition of monotone sequences and limits of monotone sequences
 - A. Monotone + Bounded $\Rightarrow$ Converges to a real number
 - B. Monotone + Unbounded $\Rightarrow$ Diverges to $\pm\infty$.

Activity

Suppose $s_n \geq 0$ for all n and $\lim s_n = s$ with $s > 0$. Prove using the ϵ, δ definition that $\lim \sqrt{s_n} = \sqrt{s}$.

Template: for proving $\lim s_n = s$

- I. “Fix $\epsilon > 0$.”
- II. “Let $N = ?$ ”
- III. “Assume $n > N$.”
- IV. Show $|s_n - s| < \epsilon$.
- V. “Since $\epsilon > 0$ was arbitrary, $\lim s_n = s$.”

fix $\epsilon > 0$. since $\lim s_n = s$, there exists N such that if $n > N$, then $|s_n - s| < \sqrt{s} \epsilon$. Assume

$n > N$. Then

$$|\sqrt{s_n} - \sqrt{s}| = \left| (\sqrt{s_n} - \sqrt{s}) \frac{\sqrt{s_n} + \sqrt{s}}{\sqrt{s_n} + \sqrt{s}} \right|$$

$$= \left| \frac{s_n - s}{\sqrt{s_n} + \sqrt{s}} \right| = \frac{|s_n - s|}{\sqrt{s_n} + \sqrt{s}}$$

$$\leq \frac{|s_n - s|}{\sqrt{s}} < \frac{\sqrt{s} \epsilon}{\sqrt{s}} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary,

$$\lim \sqrt{s_n} = \sqrt{s}.$$

Activity

Suppose $s_n > 0$ for all n . Prove that $\lim s_n = 0$ if and only if

$$\lim \frac{1}{s_n} = +\infty.$$

Template: for proving $\lim a_n = +\infty$

I. "Fix $M > 0$."

II. "Let $N = ?$ "

III. "Assume $n > N$."

IV. Show $a_n > M$.

V. "Since $M > 0$ was arbitrary, $\lim a_n = +\infty$."

($\Rightarrow$) Assume $\lim s_n = 0$. Fix $M > 0$

Since $\lim s_n = 0$, there exists

N such that: if $n > N$, then

$$|s_n - 0| < 1/M.$$

Assume $n > N$. Then $|s_n| < 1/M$.

Since $s_n > 0$, then $s_n < 1/M$.

Then $1/s_n > M$. Since $M > 0$ was arbitrary, $\lim \frac{1}{s_n} = +\infty$.

($\Leftarrow$) Assume $\lim \frac{1}{s_n} = +\infty$. Fix $\epsilon > 0$. Since $\lim \frac{1}{s_n} = +\infty$, there

exists N such that, if $n > N$, then $\frac{1}{s_n} > \frac{1}{\epsilon}$.

Assume $n > N$, then $\frac{1}{s_n} > 1/\epsilon$. Then

$s_n < \epsilon$. Since $s_n > 0$, $|s_n - 0| < \epsilon$.

Since $\epsilon > 0$ was arbitrary, $\lim s_n = 0$.

Activity

Suppose $a \geq 0$. Prove that

$$\lim a^n = \begin{cases} +\infty, & a > 1 \\ 1, & a = 1 \\ 0, & a < 1 \end{cases}$$

Hint: When $a < 1$ prove that (a^n) is bounded and monotone. Use the previous activity when $a > 1$.

Case 1: Assume $a=1$. Then $(a^n) = (1, 1, \dots)$
so $\lim a^n = 1$.

Case 2: Assume $0 \leq a < 1$. Since $a^n < 1^n = 1$ and $a^n \geq 0^n = 0$. (a^n) is bounded. Also $a^{n+1} = a \cdot a^n < a^n$ so (a^n) is decreasing.
Since bounded, monotone sequences converge

$\lim a^n = L$ for some $L \in \mathbb{R}$.

$$L = \lim a^{n+1} = \lim a \cdot a^n = a \lim a^n = aL.$$

since $0 \leq a < 1 \Rightarrow a \neq 0$.

so $\lim a^n = 0$.

Case 3: Assume $a > 1$. Then, $0 \leq \frac{1}{a} < 1$ so by the previous case,

$\lim \left(\frac{1}{a}\right)^n = 0$. By the previous activity $\lim a^n = +\infty$.

Key Ideas/Techniques – Subsequences

1. Definition of subsequences.
2. The Bolzano-Weierstrass Theorem
3. Definition of the subsequential limit set.
4. Definition of the limit inferior and the limit superior; connection to subsequential limit set.

Activity

a) Construct a sequence with exactly **five** subsequential limits.

$(1, 2, 3, 4, 5, 1, 2, 3, 4, 5, 1, 2, 3, 4, 5, \dots)$

b) Construct a sequence whose subsequential limit set is $\mathbb{N} \cup \{+\infty\}$.

$(1, 1, 2, 1, 2, 3, 1, 2, 3, 4, \dots)$

each number $1, 2, 3, \dots$ appears infinitely often so
 $(1, 1, 1, \dots)$, $(2, 2, 2, \dots)$, $(3, 3, 3, \dots)$, ... are all subsequences
and $(1, 2, 3, \dots)$ is also a subsequence.

Key Ideas/Techniques – Cauchy Sequences

1. Definition of Cauchy sequences and template for verifying a sequence is Cauchy.
2. Cauchy $\Leftrightarrow$ Convergent

Activity

Fix $\varepsilon > 0$. From a previous activity $\lim \frac{1}{4^n} = 0$.

so there exists N such that: if $n > N$, then $|\frac{1}{4^n} - 0| < 3\varepsilon$. so if $n > N$, then $\frac{1}{4^n} < 3\varepsilon$.

Assume $n, m > N$. and $n \geq m$. Then

Let $s_n = \sum_{k=0}^n \frac{\sin(k)}{4^k}$. Prove that (s_n) is Cauchy.

Template: for proving (s_n) is Cauchy

I. "Fix $\varepsilon > 0$."

II. "Let $N = ?$ "

III. "Assume $n, m > N$." [sometimes helpful to also assume $m > n$]

IV. Show $|s_n - s_m| < \varepsilon$.

V. "Since $\varepsilon > 0$ was arbitrary, (s_n) is Cauchy."

$$|s_n - s_m| = \left| \sum_{k=0}^n \frac{\sin(k)}{4^k} - \sum_{k=0}^m \frac{\sin(k)}{4^k} \right| = \left| \sum_{k=m+1}^n \frac{\sin(k)}{4^k} \right| = \sum_{k=m+1}^n \left| \frac{\sin(k)}{4^k} \right| \leq \sum_{k=m+1}^n \frac{1}{4^k} = \frac{1}{4^{m+1}} \sum_{k=0}^{n-m-1} \frac{1}{4^k} <$$

$$\frac{1}{4^{m+1}} \sum_{k=0}^{\infty} \frac{1}{4^k} = \frac{1}{4^{m+1}} \cdot \frac{1}{1 - \frac{1}{4}} = \frac{1}{4^m} \cdot \frac{1}{3} < 3\varepsilon \cdot \frac{1}{3} = \varepsilon.$$

since $\varepsilon > 0$ was arbitrary, (s_n) is Cauchy.

Chapter 1: Review

Key Ideas/Techniques

1. Induction
2. Rational and Irrational Numbers
3. Completeness Axiom, Supremum, Infimum

Activity

Prove by induction: If $a_1, a_2, \dots$ are odd numbers, then for every $n \in \mathbb{N}$ the product $a_1 a_2 \cdots a_n$ is also odd.

we prove by induction

Base case: If $n=1$ then a_1 is odd.

Induction Step: Suppose $a_1 \cdots a_n$ is odd then there exists $k \in \mathbb{Z}$ such that $a_1 a_2 \cdots a_n = 2k+1$. Since a_{n+1} is odd, there exists

$l \in \mathbb{Z}$ such that $a_{n+1} = 2l+1$. Then

$$\begin{aligned} a_1 a_2 \cdots a_{n+1} &= (a_1 a_2 \cdots a_n) (a_{n+1}) = (2k+1)(2l+1) = 4kl + 2(k+l) + 1 \\ &= 2(kl+k+l) + 1. \text{ Since } 2l+k+l \in \mathbb{Z}, \end{aligned}$$

$a_1 a_2 \cdots a_{n+1}$ is odd.

Activity

Suppose $k > 2$ is irrational and x_1, x_2 are solutions to the equation

$$x^2 + kx + 1 = 0$$

Prove x_1 and x_2 are both irrational.

Hint: What is $x_1 + x_2$ and $x_1 x_2$?

By the quadratic equation $x_1, x_2 = \frac{-k \pm \sqrt{k^2 - 4}}{2}$.

so $x_1 + x_2 = -k$. since k is irrational, and the sum of two rationals is rational, at least one of x_1 and x_2 must be irrational.

$$x_1 x_2 = \left(\frac{-k + \sqrt{k^2 - 4}}{2} \right) \left(\frac{-k - \sqrt{k^2 - 4}}{2} \right) = \frac{((-k)^2 - (\sqrt{k^2 - 4})^2)}{4} = 1$$

So $x_1 = \frac{1}{x_2}$ since at least one is irrational, both are irrational.

Activity

Compute the maximum, minimum, supremum, and infimum of the following sets:

a) $A = \{x : |x - 3| < 5\} \cap \{x : |x - 2| \geq 4\}$

b) $B = \{x : x^3 < 8\} \cup [7, +\infty)$

c) $C = \{1 + \frac{(-1)^n}{n} : n \in \mathbb{N} \text{ and } n \geq k\}$ where $k \in \mathbb{N}$

a) $A = (-2, 8) \cap ((-\infty, -2] \cup [6, \infty)) = [6, 8)$ so

$\max A$ DNE, $\min A = 6$, $\inf A = 6$ $\sup A = 8$.

b) $B = (-2, 2) \cup [7, +\infty)$ $\max B$ DNE, $\min B$ DNE, $\inf B = -2$, $\sup B = +\infty$.

c) If k is odd, $C = \{1 - \frac{1}{k}, 1 + \frac{1}{k+1}, 1 - \frac{1}{k+2}, 1 + \frac{1}{k+3}, \dots\}$ $\min C = \inf C = 1 - \frac{1}{k}$
 $\max C = \sup C = 1 + \frac{1}{k+1}$

If k is even $C = \{1 + \frac{1}{k}, 1 - \frac{1}{k+1}, 1 + \frac{1}{k+2}, 1 - \frac{1}{k+3}, \dots\}$ $\min C = \inf C = 1 - \frac{1}{k+1}$
 $\max C = \sup C = 1 + \frac{1}{k}$.